MT 1003 – Calculus and Analytical Geometry

Homework #3

Intermediate Value Theorem and Slant Asymptotes

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Instructions:

1. All homework files must follow the naming format

   `Section_RollNumber_HW3`

   (for example, `A_26I1234_HW3`). Submissions that do not follow this format will not be evaluated.
2. Write your name and roll number clearly on every page.
3. Submit your complete homework as a single scanned PDF file.
4. Show all essential calculations. Unsupported answers will receive no credit.

Failure to comply with these instructions may result in zero marks.

Section A: Intermediate Value Theorem

Q1. Locating a root. Let

$$f(x) = x^3 + x - 1.$$

(a) Verify that f satisfies the hypotheses of the Intermediate Value Theorem on $[0, 1]$.

(b) Show that the equation $x^3 + x - 1 = 0$ has at least one solution in $(0, 1)$.

Q2. Checking the hypotheses. A student argues that

$$h(x) = \frac{1}{x - 2}$$

must have a zero in $(1, 3)$ because $h(1) = -1$ and $h(3) = 1$.

(a) Explain precisely why the Intermediate Value Theorem cannot be applied on $[1, 3]$.

(b) Verify directly that $h(x)$ has no zero.

(c) State the missing hypothesis in the student's argument.

Section B: Brief Introduction to Slant Asymptotes

You have already discussed vertical and horizontal asymptotes. A **slant asymptote**, also called an **oblique asymptote**, is a nonhorizontal line

$$y = mx + b, \qquad m \neq 0,$$

such that

$$\lim_{x\to\infty} \big[f(x) - (mx + b)\big] = 0 \quad \text{or} \quad \lim_{x\to-\infty} \big[f(x) - (mx + b)\big] = 0.$$

For a rational function, a slant asymptote usually occurs when the degree of the numerator is exactly one more than the degree of the denominator. The equation of the slant asymptote has the form

$$\boxed{y = mx + c}.$$

The values of m and c are obtained by the **polynomial long-division method**. Dividing the numerator by the denominator gives

$$f(x) = mx + c + \frac{r(x)}{q(x)},$$

where the remainder term $r(x)/q(x) \to 0$ as $x \to \pm\infty$. Therefore, the graph approaches the line $y = mx + c$ for large positive or negative values of x.

Solved Example

Find the vertical and slant asymptotes of

$$f(x) = \frac{2x^2 + 3x - 1}{x - 1}.$$

Use polynomial long division:

$$\begin{array}{r} 2x + 5 \\ x - 1 \,\big)\, 2x^2 + 3x - 1 \\ \underline{-(2x^2 - 2x)} \\ 5x - 1 \\ \underline{-\,(5x - 5)} \\ 4 \end{array}$$

Thus, the quotient is $2x + 5$ and the remainder is 4, so

$$\frac{2x^2 + 3x - 1}{x - 1} = 2x + 5 + \frac{4}{x - 1}.$$

The denominator is zero at $x = 1$, while the numerator is $4 \neq 0$. Therefore, $x = 1$ is a **vertical asymptote**.
Also,

$$f(x) - (2x + 5) = \frac{4}{x - 1} \longrightarrow 0 \qquad \text{as } x \to \pm\infty.$$

Hence the **slant asymptote** is

$$\boxed{y = 2x + 5}.$$

Section C: Asymptotes

For Questions 3–5:

(i) find all vertical asymptotes;

(ii) use polynomial division to find the slant asymptote;

(iii) verify the slant asymptote using an appropriate limit;

Q3.

$$f(x) = \frac{x^2 + 3}{x - 2}.$$

Q4.

$$g(x) = \frac{3x^2 - x + 4}{x + 1}.$$

Q5.

$$h(x) = \frac{x^3 + 2x^2 + 1}{x^2 - 1}.$$

Q6. Comparison with a horizontal asymptote. Consider

$$r(x) = \frac{2x^2 + 1}{x^2 - 4}.$$

(a) Find all vertical asymptotes.

(b) Find the horizontal asymptote.

(c) Explain why this function does not have a slant asymptote.

Q7. Server-response model. The excess response time of a server is modeled by

$$R(x) = x + 4 + \frac{9}{x - 3}, \qquad x > 0,$$

where x represents the request rate in hundreds of requests per second.

(a) Identify the vertical and slant asymptotes.

(b) Determine whether the graph lies above or below its slant asymptote when $x > 3$.

(c) Determine whether the graph can intersect its slant asymptote. Justify your answer algebraically.

(d) Explain the practical warning suggested by the vertical asymptote.

Q8. Python output. Copy, paste, and run the following Python code. Attach the graph obtained as the output. From the output graph, identify the vertical asymptote and the slant asymptote.

```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(-6, 6, 2000)
y = (x**2 + 3)/(x - 2)
plt.plot(x, np.where(abs(x - 2) > 0.05, y, np.nan))
plt.plot(x, x + 2, 'r--'); plt.axvline(2, color='k', ls='--')
plt.ylim(-15, 15); plt.grid(); plt.show()
```

End of Homework #3